NLB OKR Quick Check

Documentation of the app and explanation of the measures

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Important: The app is a demo for a wellness application and not a medical device. It does not make a diagnosis. The cited studies used other methods (e.g. magnetic search coils, infrared or video oculography, fMRI, often large stimulus fields). The findings indicate which brain areas can influence a measure. They cannot be transferred directly to a single measurement with the iPad camera. Abnormal values should be clarified by a doctor (neurology, neuro-ophthalmology, ENT).

Citation style: References appear in the text in square brackets: [number, section]. The number refers to the bibliography at the end. The section names the part of the study summary (abstract) the statement comes from, for example “Results” or “Conclusions”. For unstructured abstracts it says “Abstract”. The PubMed abstracts were evaluated. The full texts were not read.

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1 The optokinetic reflex in brief

When a large, high-contrast pattern moves through the visual field, the eyes follow it automatically. This is the optokinetic reflex (OKR). The resulting eye movement is called optokinetic nystagmus (OKN). It consists of two parts [1, Abstract]:

- **Slow phase:** The eye follows the pattern smoothly in the direction of motion. This phase reveals the most about motion processing.
- **Fast phase:** A short reset jump (a saccade) against the direction of motion brings the eye back.

Over time this produces a sawtooth. Its rhythm and height vary considerably from beat to beat, even in healthy people [8, Abstract].

Look OKN and stare OKN

There are two forms of OKN, and which one occurs depends on the instruction given to the test person:

- **Look OKN:** The person deliberately tracks individual stripes. The slow phases are long and the resetting jumps are large and infrequent [2, Purpose; 3, Abstract].
- **Stare OKN:** The person looks passively straight ahead. This is more reflexive, with shorter slow phases and lower gain [2, Purpose].

In an fMRI study with 6 healthy participants, look OKN activated considerably more cortex than stare OKN, among other areas in the parietal, temporal and occipital lobes [4, Abstract]; the Brodmann assignment (roughly BA7, BA18/19, BA21, BA39/40) is not taken from the study. Stare OKN is controlled mainly by the brainstem (Mesencephalon, Pons, Medulla) and the cerebellum (Cerebellum) [12, Abstract]. The app provides a fixed instruction to be read aloud: “Look at the picture in a relaxed way. You may follow it with your eyes. Keep your head still.” This probably produces a mixed form with a look component. For comparable measurements the instruction must therefore always be the same.

2 The app

2.1 Technology

- **Web app:** The app is a web page. It runs in Safari on iPad or iPhone, ideally as an app on the home screen.
- **Camera:** The front camera delivers about 30 frames per second, up to 60 on some devices.
- **Eye detection:** The “MediaPipe Face Landmarker” program finds 478 facial points in every camera frame, including the iris centres and the eye corners.
- **Eye position:** The app calculates the position of the iris between the eye corners and divides it by the eye width. This keeps the value largely constant when the head or the device shifts in the image. The app averages both eyes.
- **Precedent:** Automatically analysed OKN measurements using eye tracking are already used in research. In a pilot study with 23 adults, the visual acuity determined in this way agreed well with the eye chart [28, Methods; 28, Results]; the authors also see potential for children who cannot yet speak [28, Conclusions].
- **Data:** All measurements stay on the device. The app does not store any video. Data are backed up by exporting them as files.

2.2 Measurement workflow

1. **Set up device:** Set the eye-to-screen distance and the device type once. Measurements are always taken with the iPad in landscape in a stand.
2. **Person and measurement conditions:** Create the test person under a pseudonym. For each person the pattern (stripes or symbols), colour (black, red or blue on white), width, two speeds and, optionally, the vertical measurement are saved.

3. **Setup check:** Read the fixed instruction aloud and start the measurement. The app checks camera, face, eyes, light, position and stillness. If no item is red for 2 s, calibration starts automatically.
4. **Calibration:** After a countdown, a rabbit appears as a gaze target in turn at centre → left → centre → right → centre, with vertical measurement also at top and bottom, each for 1.8 s. The measurement starts only when all check criteria are met (section 2.4).
5. **Measurement:** ← and → alternate, 3 rounds per speed; with vertical measurement, ↑ and ↓ follow in the same way. Each round lasts 8 s, preceded by a 3 s pause with fixation cross, direction arrow and progress indicator. Tapping pauses. The app automatically repeats invalid rounds once at the end.
6. **Result:** Three tiles: right/left asymmetry with measurement fluctuation and rating, up/down asymmetry (if measured) and gain. Individual rounds and the eye trace are shown under "Details".
7. **Progress:** The app compares only measurements of the same person under the same conditions. The first measurement is the baseline; the progress appears as charts and a table.
8. **Export:** JSON (complete, re-importable) and CSV (for Excel) via the share menu: Files, AirDrop or SeaFilePro.

2.3 Settings and defaults

Setting	Default	Meaning
Distance eye–screen	40 cm	Converts screen points into visual angle (degrees). Please measure, do not estimate. Set once in the device setup.
Device (points per inch)	iPad 132, iPad mini 163, iPhone 153	Physical size of the screen points. Needed for conversion to degrees.
Directions	← →, optionally ↑ ↓	Direction of stripe motion. Vertical stripes for horizontal motion, horizontal stripes for vertical motion.
Speeds	8 and 12 °/s	Two speeds, freely selectable; speed of the pattern in visual angle.
Duration per round	8 s	The first second is discarded because the reflex is still building up.
Pattern	Stripes	Stripes or symbols (star, moon, sun, fish, dog, rabbit, elephant, randomly mixed).
Colour	Black	Black, red or blue on a white background.
Width	8° per pair (0.125 cycles/°)	Width of one coloured plus one white stripe. For symbols, the grid spacing; a symbol is about as large as a stripe.
Order	fixed: per speed ← → alternating, 3 rounds	Distributes fatigue evenly across both directions.

2.4 Setup check and calibration

The setup check runs before calibration. Red blocks the start, yellow is allowed. If no item is red for 2 s, calibration starts automatically.

Check	Green if ...
Camera	at least 20 frames per second (yellow from 12)
Face detected	in at least 90 % of the frames of the last 1.5 s
Eyes open, iris visible	palpebral fissure at least 18 % of the eye width
Light	mean image brightness between 27 % and 90 %
Face centred	midpoint between the eyes in the central area of the camera image
Head still	Fluctuation below 3 % of the eye distance
Distance	Not checked but set up once. During the measurement the app warns if there is a clear deviation.

Calibration: all criteria below must be met before the pattern starts. Without vertical measurement, the vertical criteria do not apply.

Criterion	Limit	Purpose
Detection rate per point	≥ 60 %	Enough valid frames per gaze target
Horizontal fit (R^2)	≥ 0,8	Eye feature and gaze angle are linearly related
Horizontal separation	≥ 4 × noise	Left and right clearly distinguishable
Vertical fit (R^2)	≥ 0,7	Vertically the iris movement is smaller and harder to measure
Vertical separation	≥ 3 × noise	Up and down clearly distinguishable
Repeatability of the centre	spread ≤ 1.5° (horizontal) / 2.5° (vertical)	Fixating the centre five times gives consistent results
Head movement	≤ 25 % of the eye distance	Head and device stayed steady

2.5 Quality control during the measurement

- **Invalid frames:** Blink: the palpebral fissure is smaller than 55 % of the calibration value. Such frames count as invalid.
- **Shaking:** Face movement in the camera image of more than 1 eye distance per second. These frames are not used for the gain.
- **Distance correction:** If the face comes closer, the visual angle of the stripes becomes larger. The app therefore converts the stripe speed to the actual distance using the face size. If the distance differs by more than about 15 % from the distance during calibration, the app shows the hint “please come a little closer” or “please move back a little”.
- **Round quality:** “good”: at least 70 % valid frames and at most 20 % shaking. “fair”: at least 50 % and at most 40 %. Otherwise “invalid”. The app automatically repeats invalid rounds once at the end.

2.6 Limits of measurement accuracy

- **Frame rate:** Laboratory methods (magnetic search coil, infrared oculography) sample much faster. The app measures at about 30 frames per second. The slow phase, and thus the gain, can be captured well. Fast phases last only 20–50 ms and are therefore captured only roughly.
- **Stimulus field size:** At 40 cm the iPad covers about 30° × 22° of the visual field. Laboratory studies sometimes use much larger stimulus fields, e.g. 72° × 60° [6, Methods]. In one study, a very small central stimulus field (10° × 6°) reduced the gain and abolished the up/down asymmetry [7, Abstract]. How the iPad stimulus field affects the results has not been studied.
- **One value for both eyes:** The app averages both eyes. It cannot detect differences between the eyes, such as those described in dyslexia (section 9.1).
- **No validation data:** There are no studies yet on the test-retest reliability of the app. The app estimates the fluctuation within a measurement from the repetitions; how much the values vary from day to day has to be determined for each person (section 5).
- **Camera position:** The front camera is not located directly in front of the eyes. This can produce a small device-related asymmetry. When comparing with earlier measurements of the same person it cancels out, as long as the device and setup remain the same.

3 How the brain produces the OKR

The chain from the eye to the eye muscle, simplified:

1. **Retina:** The retina signals image motion, i.e. image slip across the retina (retinal slip).
2. **Pretectum (NOT/DTN):** The nucleus of the optic tract (NOT) and the dorsal terminal nucleus (DTN) in the pretectum (Mesencephalon), at the junction of thalamus and midbrain, contain direction-selective neurons. In monkeys, neurons in the left NOT/DTN prefer leftward motion and vice versa [13, Abstract]. In humans, fMRI during horizontal and vertical OKN found activation in the thalamo-mesencephalic junction, which includes the pretectal nuclear complex [11, Abstract].

3. **Cerebral cortex:** MT/V5 and MST, at the junction of the temporal and occipital lobes, process motion (BA19 and partly BA37). In monkeys, MT and MST project to the NOT/DTN and to the pontine nuclei [15, Abstract]. In an fMRI study with 10 healthy participants, optokinetic stimulation activated visual motion areas as well as parietal and frontal oculomotor areas, with emphasis on the right hemisphere [10, Abstract].
4. **Vestibular nuclei and velocity storage (Medulla, Pons, Cerebellum):** The vestibular nuclei combine visual and balance signals to estimate self-rotation [16, Abstract]. A velocity storage mechanism maintains eye velocity; models also use it to explain the after-nystagmus in darkness [17, Abstract].
5. **Cerebellum (Cerebellum):** The flocculus, uvula and oculomotor vermis were active during OKN in fMRI studies [11, Abstract; 12, Abstract]. In patients with cerebellar infarction, reduced slow-phase gain was associated with damage to the uvula and pyramis [18, Abstract].
6. **Saccade generators (Pons, Mesencephalon):** Like saccades, fast phases are generated by burst neurons in two centres [22, Abstract]:
    - horizontal in the paramedian pontine reticular formation (PPRF) of the pons (Pons; studies in monkeys) [21, Abstract],
    - vertical in the rostral interstitial nucleus of the medial longitudinal fasciculus (riMLF) in the midbrain (Mesencephalon) [19, Abstract].

fMRI showed activation during horizontal OKN that the authors most likely attribute to the PPRF, and during vertical OKN activation in the region of the riMLF; they describe these as first assignments [11, Abstract].

The reticular formation is a net-like network of neurons that runs through the entire brainstem (Mesencephalon, Pons, Medulla). Among other things, it controls wakefulness and attention, breathing and circulation, muscle tone, protective reflexes such as swallowing and coughing, and pain suppression. Its part in the pons, the PPRF, generates horizontal saccades and thus the rightward and leftward resetting jumps of OKN. Wakefulness also matters for the measurement: a tired or distracted child may follow the stripes less well (textbook knowledge).

7. **Eye-muscle nuclei (Pons, Mesencephalon):** Nuclei III (oculomotor nerve), IV (trochlear nerve) and VI (abducens nerve) control the external eye muscles [23, Abstract].

https://www.youtube.com/watch?v=WlTFA99sSlI&t=11s

3.1 Eye muscles by direction of movement

The following assignment is anatomical textbook knowledge. The cited papers document only individual nuclei and connections.

Eye movement	Muscles	Nerve / nucleus	Mediated by
to the right (both eyes)	right lateral rectus + left medial rectus	right VI (Pons), left III (Mesencephalon)	Abducens interneurons (Pons) via the medial longitudinal fasciculus (MLF) (textbook knowledge; on the MLF: [24, Abstract])
to the left (both eyes)	left lateral rectus + right medial rectus	left VI (Pons), right III (Mesencephalon)	as above, mirrored
upward	superior rectus + inferior oblique	III (Mesencephalon)	riMLF, interstitial nucleus of Cajal (Mesencephalon) [24, Abstract]
downward	inferior rectus + superior oblique	III and IV (Mesencephalon)	riMLF (Mesencephalon) [19, Abstract]

When the stripes move to the right, both eyes move to the right in the slow phase. The resetting jump goes to the left. Horizontal saccades are generated by the PPRF (Pons) on the side towards which they are directed (textbook knowledge).

4 The measures

For each measure the following are given: definition, calculation in the app, normal values, meaning, brain areas and eye muscles involved, and notes on comparing several measurements. What an abnormality may mean anatomically is described in section 5.4; findings in neurodevelopmental disorders are in chapter 9. In the result, the app shows only the gain and the two asymmetries.

4.1 Gain (slow-phase gain)

Definition: Eye velocity in the slow phase divided by the stripe speed. 1.0 means the eye moves exactly as fast as the stripes. 0.5 means half as fast.

Simply put (memory aid): The gain is the eyes' "keep-up rate". Imagine the stripes as a train passing a platform and the eyes as someone running alongside it. At gain 1.0 the runner is exactly as fast as the train. At 0.8 they manage 80 % of the speed, slowly fall behind and then jump forward again (the resetting jump). At 0.5 they manage only half the speed and have to jump back often.

Example: The stripes move at 10 °/s and the eyes manage 8 °/s: gain = 8 ÷ 10 = 0.8. In healthy people the gain is usually somewhat below 1; in one study it was always below 0.85 [5, Abstract].

How the app calculates it: Eye position in degrees (via calibration) → velocity from frame to frame → remove fast phases and shaking → median of the remaining velocities ÷ distance-corrected stripe speed. For each round, i.e. each direction, speed and repetition. In the result the app shows the mean over all valid rounds and, below it, the gain per direction.

Normal values and influencing factors

- **Healthy adults:** In a study with 7 healthy participants (stare instruction, large-field dot pattern) the gain was always below 0.85 and decreased with increasing speed [5, Abstract].
- **Age:** In a cross-sectional study (monocular testing, small stimulus field) the gain rose in the 2nd and 3rd year of life, remained stable until about age 50 and then fell by 6–18 % [9, Results].
- **Stimulus:** For vertical OKN, the gain was highest at low speed and with coarse stripes [6, Results]. Another study, by contrast, found higher eye velocities with finer stripes [8, Abstract]. Therefore never change the stripe width for comparisons.
- **Binocular:** In one study, binocular viewing produced a higher gain than monocular viewing [5, Abstract].
- **Instruction:** Look OKN has a higher gain than stare OKN [2, Purpose]. The fast phases are larger and less frequent with look OKN, and smaller and more frequent with stare OKN [3, Abstract].

Meaning

- **General:** The gain shows how well the brain detects image motion and converts it into an equally fast eye movement. The goal is a stable retinal image.
- **For the app:** Because of the smaller stimulus field and the lower frame rate, the values are not directly comparable with laboratory studies (section 2.6).

Brain areas and eye muscles involved

- **Input:** Retina → NOT/DTN (Mesencephalon) [13, Abstract] and cortical motion areas MT/MST (BA19/37) [15, Abstract].

- **Processing:** Vestibular nuclei (Medulla, Pons) and velocity storage (Medulla, Pons, Cerebellum) [16, Abstract], cerebellum (Cerebellum: uvula, flocculus, vermis) [11, Abstract; 12, Abstract].
- **Execution:** horizontal: lateral and medial rectus (III, VI); vertical: superior/inferior rectus and the oblique muscles (III, IV).

Comparing two or more measurements

- **Same person:** If the gain falls evenly in all directions, this is often due to attention, fatigue, light or a different distance. Only if it is confirmed on repetition should a general change in the slow tracking system be considered (cerebellum/Cerebellum, age).
- **Different people:** The app does not compare people with each other. Anyone who compares values of different people themselves should only use people of similar age [9, Results].
- **Display in the app:** The result shows the change from the baseline; the person's progress view shows the gain as a chart over all measurements.

4.2 Gain across speeds

Definition: How much the gain falls from the lower to the higher speed (default 8 and 12 °/s).

How the app calculates it: Not a separate value. The gain values per speed are shown in the result under "Details".

Normal values and influencing factors

- **Normal:** The gain decreases with increasing speed [5, Abstract]. Vertically it drops more steeply than horizontally [5, Abstract].
- **Cause:** Presumably, above a certain speed the tracking system can no longer keep up fully (interpretation).

Meaning

- **General:** The progression can be read as an indication of the reserve capacity of the tracking system. A steeper drop than in earlier measurements could mean that fast stimuli are processed less well (interpretation, not studied for the app). At 8 and 12 °/s the speeds are close together; a difference is therefore likely to be small.

Brain areas and eye muscles involved

- **Involved:** For the rapid rise of OKN, the authors of a monkey study assume a direct pathway to the pontine nuclei (Pons) and flocculus (Cerebellum) [14, Abstract].
- **Measurement in humans:** fMRI found speed-dependent activation in the pons (Pons: NRTP, PPRF) and in the uvula, flocculus and cerebellar tonsils (Cerebellum) [12, Abstract].

Comparing two or more measurements

- **Same person:** If the gain drops more than before only at the higher speed, this points to a change with fast stimuli rather than a general change (interpretation).

4.3 Right/left asymmetry

Definition: (gain right − gain left) ÷ mean of both × 100 %. Positive: patterns moving to the right are tracked better.

Simply put: The keep-up rate is worse in one direction than in the other, for example 0.6 to the right and 0.8 to the left. In the train picture: when the train goes to the right, the eyes keep up less well.

How the app calculates it: For each round, the pair ← and → at the same speed gives one asymmetry value, i.e. up to 6 values with 2 speeds × 3 rounds. The app shows their mean and, as measurement fluctuation, the 95 % confidence interval (t-distribution). Invalid rounds are excluded.

Rating in the app: “No clear direction difference” below 10 %; “clear” from 10 % if the confidence interval does not include zero; otherwise “unclear”. The 10 % threshold is a setting of the app, not a value from the literature. In the case of a clear difference, the result text reads: “This pattern matches lesion studies in which OKN towards the side of right-sided (or left-sided) brain damage was weaker. Whether this person has the same cause cannot be shown by the measurement. Small differences can also be caused by the device.” For the lesion studies see section 5.4.

Normal values and influencing factors

- **Normal:** Horizontally there is no group asymmetry in healthy people [5, Abstract; 2, Results]. According to a review, horizontal OKN is symmetrical in healthy adults [1, Abstract].
- **Infants:** With monocular testing, an asymmetry was present in the first 5 months of life; only at 11 months was OKN symmetrical in all children examined [9, Results]. The app tests binocularly.

Meaning

- **General:** A clear and reproducible asymmetry can be a clinically relevant finding in horizontal OKN. It may indicate a one-sided disturbance of motion processing or gaze control, because horizontal OKN is symmetrical in healthy adults [1, Abstract]. Localisation hints are given in section 5.4.

Brain areas and eye muscles involved

- **Pretectum (Mesencephalon):** After damage to the NOT/DTN, OKN with slow phase towards the damaged side was reduced. The opposite direction was unaffected [14, Abstract] (study in monkeys).
- **Muscles:** If the asymmetry is muscular, the lateral rectus (VI) or medial rectus (III) of one side is affected. This cannot be distinguished using the average of both eyes.

Comparing two or more measurements

- **Same person:** An asymmetry that appears newly or increases markedly may be particularly noteworthy. It should be confirmed by repeated measurements with the same setup.
- **Different people:** The app does not compare people with each other; there are no normative values for the app.

4.4 Up/down asymmetry

Definition: (gain up − gain down) ÷ mean of both × 100 %. Positive: patterns moving upwards are tracked better.

How the app calculates it: as in section 4.3, with the directions ↑ and ↓ (only with vertical measurement). The app does not rate the value as abnormal or normal but shows the change from the baseline.

Normal values and influencing factors

- **Mixed evidence:** Three studies found an upward advantage: on average about 0.15 gain [5, Abstract], in most measurements [6, Results] and at 30–70 °/s [7, Abstract]. Another found no vertical asymmetry at group level [2, Results].
- **Individual trait:** Vertical asymmetries differ between individuals but are consistent within the same person across different measurement conditions (left/right eye, look/stare) [2, Results; 2, Conclusions].
- **Stimulus field:** In one study, the up/down asymmetry disappeared when only the centre of the visual field was stimulated [7, Abstract]. On the iPad it may therefore be smaller than in the laboratory.

Meaning

- **General:** Because of the large individual differences, the vertical value is suited mainly to tracking change within the same person, less to comparisons between people.

Brain areas and eye muscles involved

- **Vertical gaze centres:** riMLF, interstitial nucleus of Cajal and posterior commissure in the midbrain (Mesencephalon) [24, Abstract; 11, Abstract].
- **Muscles:** upward: superior rectus + inferior oblique (III); downward: inferior rectus (III) + superior oblique (IV).

Comparing two or more measurements

- **Same person:** Because the personal vertical asymmetry was consistent across measurement conditions in one study [2, Conclusions], a change relative to the person's own earlier values is probably more informative than the value itself.
- **Different people:** The app does not compare people with each other; differences between people in the vertical value are often normal.

4.5 Technical parameters

These values describe measurement quality, not brain function:

Value	Meaning	What to do if values are poor
Eyes detected (%)	Proportion of valid frames in the round	Light, glasses, camera at eye level
Shaking (%)	Share of frames with device or head movement	Use a stand, support the head
Measurement fluctuation (±)	95 % confidence interval of the asymmetry from the repetitions	If the fluctuation is large, repeat the measurement; make sure the person is calm and attentive
Quality	good / fair / invalid (section 2.5)	The app repeats invalid rounds automatically
Calibration R^2 / separation	Quality of the conversion to degrees	Recalibrate if values are borderline

5 Comparing two or more measurements

5.1 What the app shows

- **Baseline:** The first measurement of a person under given conditions automatically becomes the baseline.
- **Result:** Right/left and up/down asymmetry as well as gain, each with the change from the baseline.
- **Progress:** Charts for right/left asymmetry, gain and (if measured) up/down asymmetry over all measurements of the person, plus a table.
- **Same conditions:** Measurements with a different pattern, colour, width or different speeds are shown separately and are not compared.
- **No comparisons between people:** The app does not compare people with each other and uses no normative values.

5.2 When is a difference meaningful?

There are no studies yet on the measurement accuracy of the app. Until then, the following practical procedure applies. It is a recommendation, not a literature value:

1. **Baseline:** The app estimates the fluctuation within a measurement from the 3 repetitions per speed and shows it as "±". How much the values vary from day to day only becomes clear from 2–3 baseline measurements on different days.
2. **Real change:** A later difference only counts as real if it is clearly larger than this variability and is confirmed on repetition.
3. **Pattern over single value:** A difference affecting only one direction is more meaningful than an even change across all directions. Even changes often have external causes such as fatigue, light or distance.
4. **Quality:** Check quality and shaking of the compared rounds. A "fair" round can create a spurious difference.

5.3 Patterns of change and possible meaning

The table shows which patterns were associated with which brain areas in studies. It is a guide, not a diagnostic manual.

Pattern in the comparison	Possible meaning	Brain area / eye muscle	Source
Gain evenly lower in all directions	Attention, fatigue, setup; age; general weakness of the tracking system	Cerebellum (Cerebellum: uvula, pyramis); age-related	[12, Abstract], [18, Abstract], [9, Results]
One horizontal direction lower (new R/L asymmetry)	one-sided disturbance of motion processing or tracking control	NOT/DTN (Mesencephalon) on the side towards which the weaker slow phase points (animal study); frontal eye field (BA8/6) / posterior parietal lobe (BA7, BA39/40); hemisphere	[14, Abstract], [26, Abstract], [25, Abstract]
Vertical gain lower, resetting jumps infrequent or slowed	impaired vertical saccade system	Midbrain (Mesencephalon): riMLF; muscles III/IV	[19, Abstract], [20, Abstract]
Horizontal reset jumps missing with preserved gain	impaired horizontal saccade generation	Pons (Pons): PPRF	[21, Abstract]
Gain dropping more at the higher speed	May indicate a smaller reserve of the fast tracking system	Pontine nuclei (Pons), flocculus, uvula (Cerebellum)	[12, Abstract]
Gain higher than before	e.g. look instead of stare instruction	more cortical involvement with look OKN	[2, Purpose], [4, Abstract]

5.4 Localisation hints: what an abnormality may mean anatomically

The following findings come from lesion studies in patients and animals. They show which brain areas can produce a particular OKN abnormality. The anatomical relationships are not tied to a specific disease; whether they apply in the same way to neurodevelopmental disorders has not been studied. An abnormality in the app therefore does not show where a disorder is located, but where a more detailed examination could start.

- **One horizontal direction weaker:** In monkeys, lesions of the NOT/DTN (Mesencephalon) reduced OKN with slow phase towards the side of the lesion; the opposite direction remained unaffected [14, Abstract]. In patients with lesions in the region of the frontal eye field or the posterior parietal lobe, pursuit towards the side of the lesion was slowed, with optokinetic stimulation in 8 of 13 cases with parietal lobe lesions [26, Abstract]. In neglect after right-sided brain damage, the velocity, amplitude and frequency of the slow phases moving to the right were reduced [27, Abstract].
- **Hemisphere:** With damage to the entire left hemisphere, a directional asymmetry with reduced slow-phase velocity was found; with lesions of the parietal lobe alone there was no directional preponderance [25, Abstract]. Which direction was affected cannot be read unambiguously from the abstract.

- **Vertical resetting jumps impaired:** Bilateral damage to the riMLF in the midbrain (Mesencephalon) impaired fast vertical eye movements [19, Abstract]. In a case report, even unilateral riMLF damage led to upward and downward gaze palsy [20, Abstract].
- **Horizontal resetting jumps impaired:** In monkeys, inactivation in the region of the PPRF (Pons) markedly slowed horizontal saccades [21, Abstract]. An isolated disturbance of horizontal saccades points to a lesion in the pons [24, Abstract].
- **Gain reduced overall:** In patients with cerebellar infarction, reduced gain of the slow OKN phase was associated with damage to the uvula and pyramis (Cerebellum) [18, Abstract].
- **One eye different from the other:** With a lesion of the medial longitudinal fasciculus (MLF), adduction of the eye on the side of the lesion is impaired (internuclear ophthalmoplegia) [24, Abstract]. Differences between the eyes therefore point to the brainstem, ocular motor nerves or eye muscles rather than to one hemisphere (textbook knowledge). The current app averages both eyes and cannot detect such differences.

5.5 Comparing the same person over time

- **Strength:** This is probably the most informative application. In one study, personal characteristics such as the vertical asymmetry were consistent within the same person across different measurement conditions [2, Conclusions]; whether this also holds over weeks has not been shown.
- **Age effect:** In a cross-sectional study, the gain rose in the 2nd and 3rd year of life and fell slightly from about age 50 [9, Results]. In toddlers and older adults, age-related changes are therefore to be expected over longer periods.
- **Requirements:** Same device, same distance, same measurement conditions (pattern, colour, width, speeds), same instruction, similar time of day, same light.

5.6 Comparing different people

- **Age:** The app does not compare people with each other. Anyone who compares values of different people themselves should only use people of similar age [9, Results].
- **Asymmetries:** Vertical asymmetries differ between individuals [2, Conclusions]. Horizontal OKN is symmetrical in healthy adults [1, Abstract].

6 Recommendations for good measurements

- **Setup:** iPad in landscape in a stand, camera at eye level, distance measured with a tape measure.
- **Light:** even, from the front, no backlight from a window.
- **Glasses:** reflective lenses make detection harder. For comparisons always do the same (with or without).
- **Instruction:** read out the instruction shown in the app word for word every time.
- **Measurement conditions:** do not change pattern, colour, width and speeds within a measurement series; the app shows measurements with different conditions separately.
- **Repetition:** the app repeats invalid rounds automatically. At the start, take several baseline measurements on different days.
- **Backup:** export after every session, for example via AirDrop or to SeaFile.

7 Glossary

Term	Explanation
OKR / OKN	optokinetic reflex / optokinetic nystagmus: tracking movement of the eyes in response to a moving pattern
Slow phase	Tracking movement in the direction of motion
Fast phase	Reset jump (saccade) against the direction of motion

Term	Explanation
Gain	The eyes' "keep-up rate": eye velocity ÷ pattern velocity (1.0 = eyes exactly as fast as the pattern)
Symmetry	The eyes follow the pattern equally well in both directions (e.g. the same gain to the right and to the left)
Asymmetry	The eyes follow less well in one direction than in the other, given in % (e.g. 0.6 to the right and 0.8 to the left)
Look / stare	active tracking / passive looking
NOT / DTN	nucleus of the optic tract / dorsal terminal nucleus: entry station of the OKR in the pretectum (Mesencephalon)
MT/V5, MST	Motion areas of the cerebral cortex (BA19/37)
PPRF	paramedian pontine reticular formation: centre for horizontal saccades (Pons)
Reticular formation	net-like network of neurons in the brainstem (Mesencephalon, Pons, Medulla): wakefulness, breathing and circulation, muscle tone, reflexes, pain suppression; contains the PPRF for horizontal saccades
riMLF	rostral interstitial nucleus of the medial longitudinal fasciculus: centre for vertical saccades (midbrain, Mesencephalon)
MLF	medial longitudinal fasciculus: connects the abducens (Pons) and oculomotor nuclei (Mesencephalon)
Flocculus, uvula, vermis	Parts of the cerebellum (Cerebellum) that fine-tune eye movements
Velocity storage	Velocity storage in the vestibular nuclei (Medulla, Pons) and cerebellum (Cerebellum)
OKAN	optokinetic after-nystagmus: continuation of the nystagmus in darkness
Round	a measurement of 8 s in one direction at one speed
Baseline	first measurement of a person under given conditions; reference for all later measurements
Measurement fluctuation	95 % confidence interval of the asymmetry from the repetitions within one measurement

8 Before-and-after measurement with training

A measurement before and after eye training in the same person is probably the most useful application of the app. This applies to children and adults. Each person serves as their own reference, and the large differences between people play no role. For a change to mean something, the following points are important.

8.1 Know the personal variability

Values vary from day to day even without training, especially in children. How much this applies to the app has not yet been studied.

1. **Baseline measurements:** Before the training, take 2–3 baseline measurements on different days, e.g. within 1–2 weeks. These measurements show the person's normal variability.
2. **Threshold:** A change after the training only counts if it is clearly larger than this variability. Practical rule of thumb: more than twice the range of the baseline measurements. This is a recommendation, not a literature value.
3. **Follow-up measurements:** Also take 2 measurements after the training. A single measurement can be an outlier.

Time point	Measurements	Purpose
Weeks 1–2, before the training	2–3, on different days	determine personal baseline variability
Training phase	optional: 1 interim measurement	see the trend
After the training	2, on different days	test the change against the baseline variability
A few weeks later (optional)	1–2	check whether a change persists

8.2 Keep everything the same

- **Setup:** Device, distance, stand, light and time of day.
- **Measurement conditions:** pattern, colour, width and speeds.
- **Instruction:** word for word the same. In children this is probably an important confounding factor: a child who looks more attentively the second time has a higher gain without anything having changed in the brain. Look OKN has a higher gain than stare OKN [2, Purpose].

8.3 What the measurement cannot prove

If a value changes, there can be many reasons, not just the training:

- **Familiarity with the test:** The person knows the procedure and is more relaxed.
- **Attention and motivation:** Form on the day, fatigue, interest.
- **Normal development in children:** In a cross-sectional study, OKN gain rose in the 2nd and 3rd year of life and then remained stable until about age 50 [9, Results]. Over a few weeks a developmental effect in school children is therefore likely to be small, but it is possible in toddlers.
- **Regression to the mean:** Someone who happened to do poorly in the first measurement often does better the next time, even without training.
- **Training transfer:** If the eye training resembles the stripe or symbol band, it may mainly be the test task that is being practised. A general improvement is then not demonstrated.

Important: The measurement can show that something has changed in this person. On its own it does not prove that the training was the cause. That would require a comparison group without training or a single-case design: training starts at different times for several people. If the values improve only after training begins in each case, that is much stronger evidence for the training.

8.4 Which measures are most meaningful

- **Best suited, as a provisional practical choice:** Changes in asymmetry (right/left, up/down) and in the gain at the higher speed. They presumably depend less on attention than the overall gain; this has not been demonstrated.
- **Less suitable:** Beat frequency and amplitude often change simply because the person looks differently [3, Abstract]. They are less suitable as a measure of success and are not displayed by the app.
- **Quality:** Check “Eyes detected” and “Shaking” for every measurement. Only compare rounds of good quality.

8.5 Framework

- **Consent and data protection:** For children, parental consent is required. Use pseudonyms only. The data stay on the device.
- **Wording:** Word results cautiously, especially in publications. Appropriate: “In person X, the gain rose after the training beyond their own baseline variability.” Not appropriate: “The training improves OKN.”

9 Evidence: neurodevelopmental disorders

PubMed was searched for OKN in connection with ADHD, autism, dyslexia, tics/Tourette, coordination disorders and genetic syndromes (as of October 2026). The search found no hits for coordination disorders. The abstracts were evaluated; the search is not systematic. The figures “n = … vs. …” give the size of the patient and comparison groups.

Key message: There is no established OKN feature for any of these disorders. The findings come from few, small and partly old studies. The app therefore cannot assign a measurement to a diagnosis. It is most suitable for

tracking the same person over time (chapter 8).

Disorder	Evidence	Findings
Dyslexia	weak to moderate	lower OKN gain; larger difference between the two eyes (one study each)
Autism	weak, contradictory	lower gain with more autistic traits; vestibular reflexes normal
ADHD	very weak	OKN matures more slowly (one study)
Tics / Tourette	case reports only	OKN once normal, once interrupted by saccades
Down syndrome (for comparison)	moderate	gain reduced depending on speed, fewer beats

9.1 Dyslexia

- **Lower gain:** Children with dyslexia had a lower gain in pursuit and OKN (n = 30 vs. 20, video-nystagmography) [29, Results].
- **Difference between the eyes:** The OKR was reduced only in the right eye, which increased the difference between the eyes. Saccades were slower, less accurate and started later (n = 40 vs. 40) [30, Results]. The authors stress that a causal relationship has not been demonstrated [30, Conclusions].
- **Unsteady gaze:** In front of a moving optokinetic background, gaze became more unstable than in controls [31, Abstract]. Only children with dyslexia and signs of mild cerebellar dysfunction were examined; fixation stability was measured, not OKN gain.
- **Meta-analysis:** A meta-analysis (13 studies) found differences in the number of saccades and fixation duration [32, Results]; the abstract makes no statement about OKN.
- **Caution:** The study on a "cerebellar-vestibular dysfunction" in learning disabilities comes from a single author and itself calls for independent confirmation in larger samples with typically developing controls [33, Abstract].

9.2 Autism

- **Gain and autistic traits:** In healthy adults, the higher the autism score (Autism-spectrum Quotient), the lower the slow-phase gain. There was no relationship for the fast phases [34, Abstract]. The people examined had no autism diagnosis; this is an association (correlation), not a diagnostic test.
- **Unusual behaviour during the test:** In 31 of 34 children with autism, OKN was abnormal: delayed onset, short duration, looking away or stereotyped behaviour [35, Abstract]. Whether the reflex itself or cooperation with the test was affected is left open by the abstract.
- **No clear difference reported:** A larger vision examination (n = 51 vs. 44) also included the optokinetic response. The abstract highlights only convergence as abnormal [36, Abstract].
- **Brainstem and cerebellum intact:** Reflexes of the balance system and saccade velocity were normal; pursuit showed only a trend towards lower gain (n = 79 vs. 62). The authors interpret the normal vestibular reflexes and the normal saccade velocity and accuracy as an indication of functionally intact pathways in the brainstem (Pons, Medulla) and cerebellum (Cerebellum), and locate the differences rather in the cerebral cortex [37, Abstract]. This study did not measure OKN itself.

9.3 ADHD

- **Slower maturation:** In 148 vs. 148 boys (7–14 years), the slow-phase OKN velocity was lower and its development delayed [38, Results]. Only the abstract is in English; some figures in it appear implausible, and it describes the OKN of healthy boys partly as stable and partly as developing.
- **Possible problem:** Inattention can lower the gain independently of any brain characteristic. In ADHD the two can therefore probably hardly be separated (own assessment, not examined in the studies found).

9.4 Tics and Tourette syndrome

- **Case report 1:** A girl with eye tics had normal OKN [39, Abstract].
- **Case report 2:** In a patient with Tourette syndrome, intrusive saccades interrupted pursuit and OKN [40, Abstract].
- **Research focus:** One study refers to earlier saccade studies with disturbances of saccade timing and inhibition; it itself found less frequent and smaller microsaccades during fixation (n = 14) [41, Abstract]. A review describes different eye movement patterns in autism, ADHD and Tourette syndrome [42, Recent Findings].

9.5 For comparison: Down syndrome

- **Clearest finding:** OKN gain fell more steeply with increasing speed than in controls, and there were fewer nystagmus beats (n = 32 vs. 32, 14–36 years) [43, Abstract].
- **Additional finding:** In 18 children with Down syndrome and additional infantile nystagmus (retrospective analysis), all had abnormal gain values in at least one test (pursuit, saccades or OKN) [44, Methods; 44, Results]. The authors suspect dysfunction of the cerebellum (Cerebellum) and/or downstream oculomotor circuits as the cause [44, Conclusions]. Because of the nystagmus, the finding cannot be generalised to all people with Down syndrome.

9.6 Implications for the app

1. **No diagnosis:** No study shows an OKN feature reliable enough to assign a diagnosis.
2. **Group comparisons:** Possible outside the app, but only with age-matched controls and the same instruction. In children, attention probably has a strong influence on the result.
3. **Difference between the eyes:** The dyslexia finding of a difference between the eyes [30, Results] cannot be measured with the current app because it averages both eyes.
4. **Autism:** For the autism question, slow-phase gain would be the obvious measure [34, Abstract]. The instruction (look or stare) should be fixed and documented; the abstract does not state which instruction the study used.

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All papers are listed in PubMed (https://pubmed.ncbi.nlm.nih.gov/<PMID>). The section references in the text refer to the PubMed abstract.

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